Practical 02: Programming Resources: Colors, Themes, Strings, Dimensions and Drawables

Source: Android Programming Practical(1).pdf | Source pages: 14–28

Aim

To develop and test an Android application demonstrating programming resources: colors, themes, strings, dimensions and drawables.

Steps to Perform the Practical

1. Create a project named Practical 2R using Empty Views Activity and Java.

2. Create or update the values resource files colors.xml, dimens.xml, strings.xml and themes.xml using the supplied code.

3. Create the activity layouts and Java activity classes shown in the practical.

4. Copy the supplied image file into app/src/main/res/drawable. Use a valid lowercase filename and keep the XML drawable reference consistent.

5. Register the activities in AndroidManifest.xml and check all resource names match the references in the layouts.

6. Run the app. Open Color Demo, Theme Demo, String Demo, Drawable Demo and Dimension Demo, returning to the main page between tests.

Result

The application is run and the expected behavior is verified according to the practical instructions and code.

Note: Use the package name and resource names from the source code consistently. If Android Studio reports a missing resource or class, check spelling and ensure all files are in the correct folders.

b.Develop an Android application demonstrating the use of Programming Resources (colors.xml, themes.xml, strings.xml, dimens.xml, drawable resources) and apply them in UI components.

Programming Resources

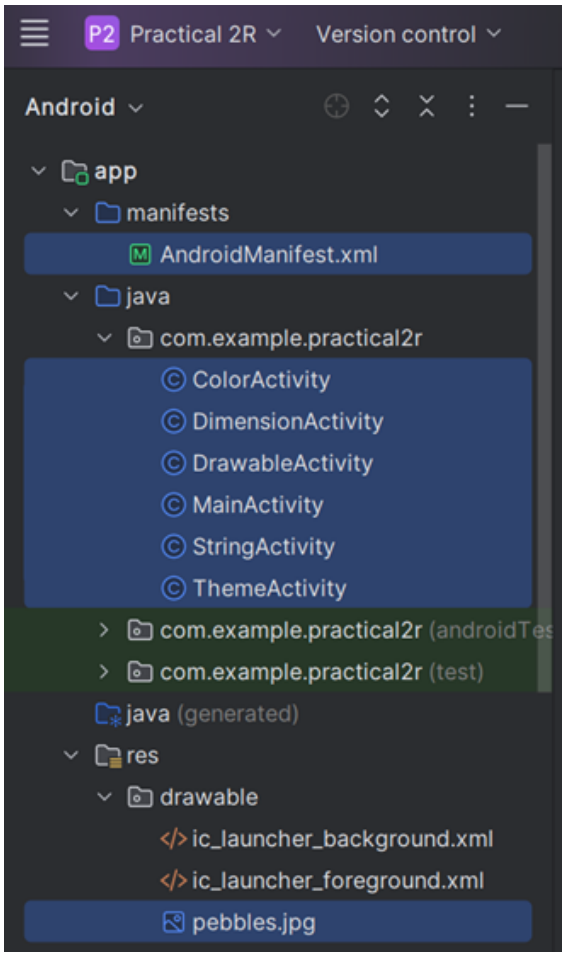
Android Resources: (Color, Theme, String, Drawable, Dimension, Image)

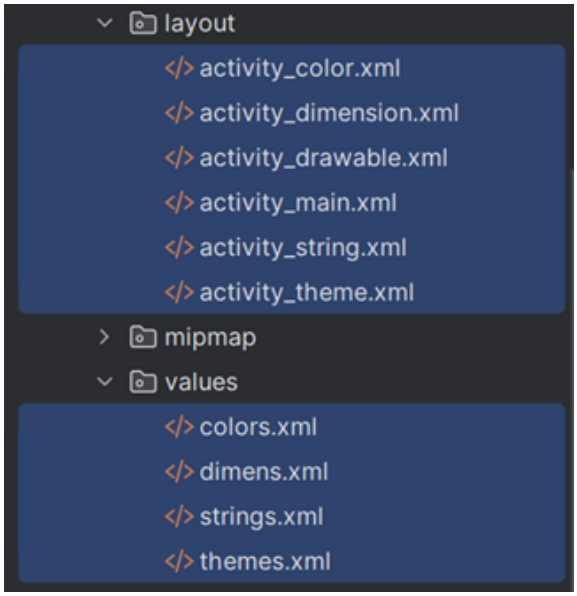
Project Name: Practical 2R

Template: Empty Views Activity

Language: Java

Project Architecture: 1 AndroidManifest.xml file, 6 Java Files,1 image in drawable image file 6 layout .xml files and 4 values .xml files: Total 18 files.





Filename: AndroidManifest.xml

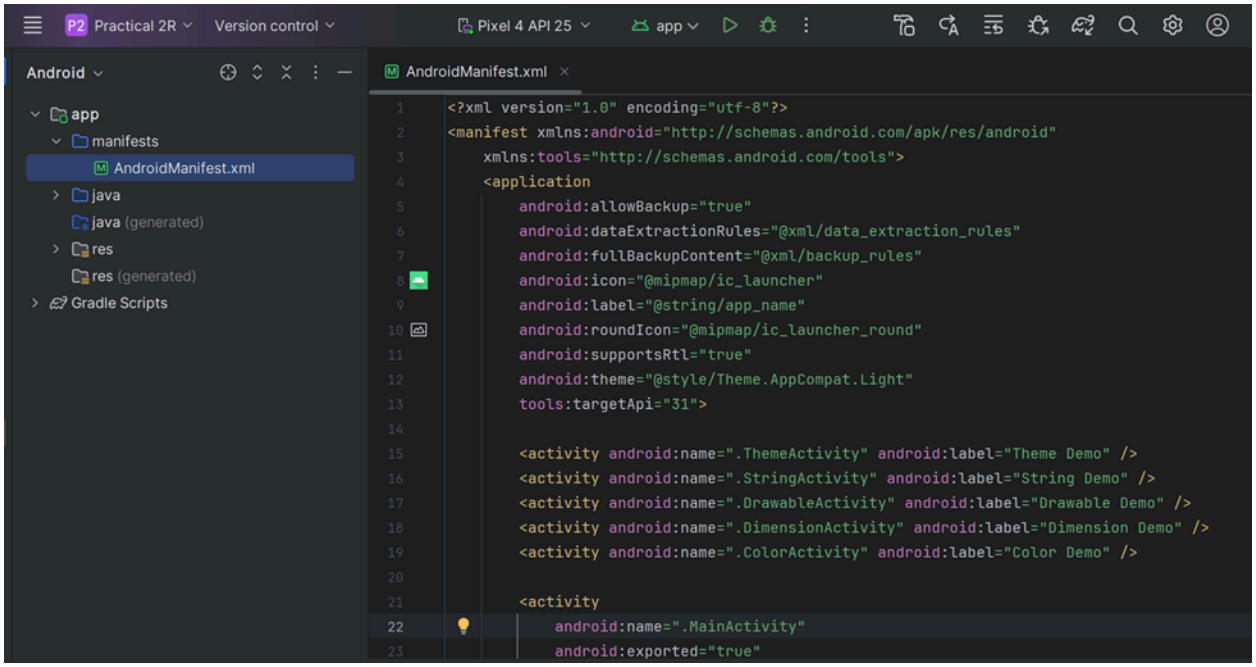
```
<?xml version="1.0" encoding="utf-8"?>
 <manifest xmlns:android="http://schemas.android.com/apk/res/android"
      xmlns:tools="http://schemas.android.com/tools">
      <application
      android:allowBackup="true"
      android:dataExtractionRules="@xml/data_extraction_rules"
      android:fullBackupContent="@xml/backup_rules"
      android:icon="@mipmap/ic_launcher"
      android:label="@string/app_name"
      android:roundIcon="@mipmap/ic_launcher_round"
      android:supportsRtl="true"
      android:theme="@style/Theme.AppCompat.Light"
      tools:targetApi="31">

      <activity android:name=".ThemeActivity" android:label="Theme Demo" />
      <activity android:name=".StringActivity" android:label="String Demo" />
      <activity android:name=".DrawableActivity" android:label="Drawable Demo"
/>
      <activity android:name=".DimensionActivity" android:label="Dimension
Demo" />
      <activity android:name=".ColorActivity" android:label="Color Demo" />

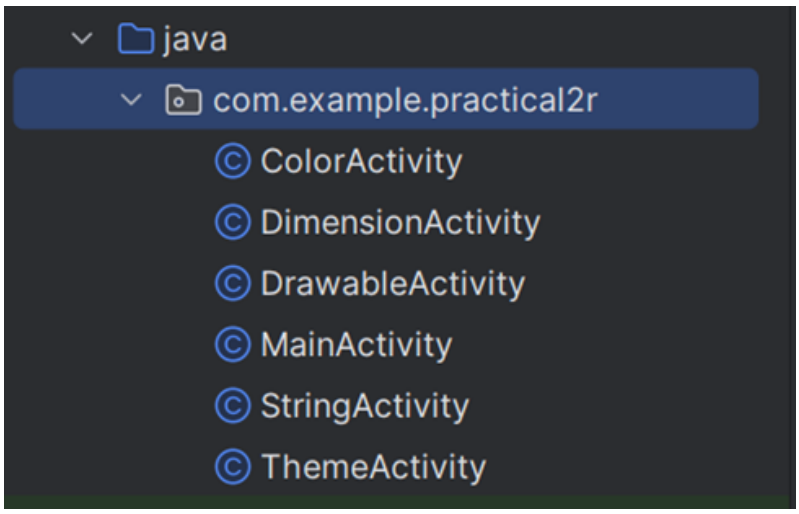
      <activity
            android:name=".MainActivity"
            android:exported="true"
            android:label="@string/app_name"
            android:theme="@style/Theme.AppCompat.Light">
            <intent-filter>
                  <action android:name="android.intent.action.MAIN" />
```

```
                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>
        </application>
</manifest>
```

Path: app>manifests>AndroidManifest.xml



Total 6 .java files path



1.Filename: ColorActivity.java

```
package com.example.practical2r;

import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;

public class ColorActivity extends AppCompatActivity {
     @Override
     protected void onCreate(Bundle savedInstanceState) {
     super.onCreate(savedInstanceState);
     setContentView(R.layout.activity_color);
     }
}
```

2.Filename: DimensionActivity.java

```
package com.example.practical2r;

import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;

public class DimensionActivity extends AppCompatActivity {
     @Override
     protected void onCreate(Bundle savedInstanceState) {
     super.onCreate(savedInstanceState);
     setContentView(R.layout.activity_dimension);
     }
}
```

3.Filename: DrawableActivity.java

```
package com.example.practical2r;

import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;

public class DrawableActivity extends AppCompatActivity {
     @Override
     protected void onCreate(Bundle savedInstanceState) {
     super.onCreate(savedInstanceState);
     setContentView(R.layout.activity_drawable);
     }
}
```

4.Filename: MainActivity.java

```
package com.example.practical2r;

 import android.content.Intent;
 import android.os.Bundle;
 import android.view.View;
 import androidx.appcompat.app.AppCompatActivity;

 public class MainActivity extends AppCompatActivity {
      @Override
      protected void onCreate(Bundle savedInstanceState) {
      super.onCreate(savedInstanceState);
      setContentView(R.layout.activity_main);
      }
      public void openColorActivity(View view) {
      startActivity(new Intent(this, ColorActivity.class));
      }
      public void openThemeActivity(View view) {
      startActivity(new Intent(this, ThemeActivity.class));
      }
      public void openStringActivity(View view) {
      startActivity(new Intent(this, StringActivity.class));
      }
      public void openDrawableActivity(View view) {
      startActivity(new Intent(this, DrawableActivity.class));
      }
      public void openDimensionActivity(View view) {
      startActivity(new Intent(this, DimensionActivity.class));
      }
 }
```

5.Filename: StringActivity.java

```
package com.example.practical2r;

 import android.os.Bundle;
 import androidx.appcompat.app.AppCompatActivity;

 public class StringActivity extends AppCompatActivity {
      @Override
      protected void onCreate(Bundle savedInstanceState) {
      super.onCreate(savedInstanceState);
      setContentView(R.layout.activity_string);
      }
 }
```

6.Filename: ThemeActivity.java

```
package com.example.practical2r;

import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;

public class ThemeActivity extends AppCompatActivity {
     @Override
     protected void onCreate(Bundle savedInstanceState) {
     super.onCreate(savedInstanceState);
     setContentView(R.layout.activity_theme);
     }
}
```

Image File Path

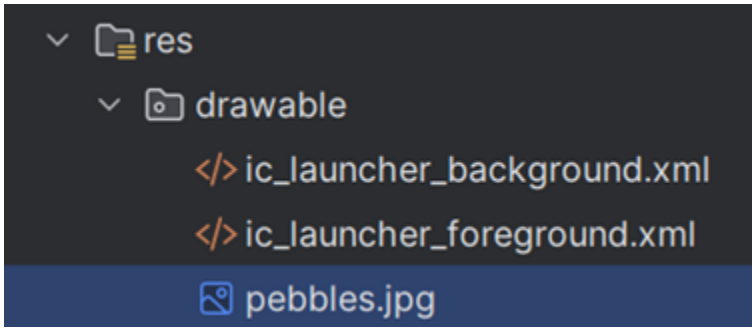
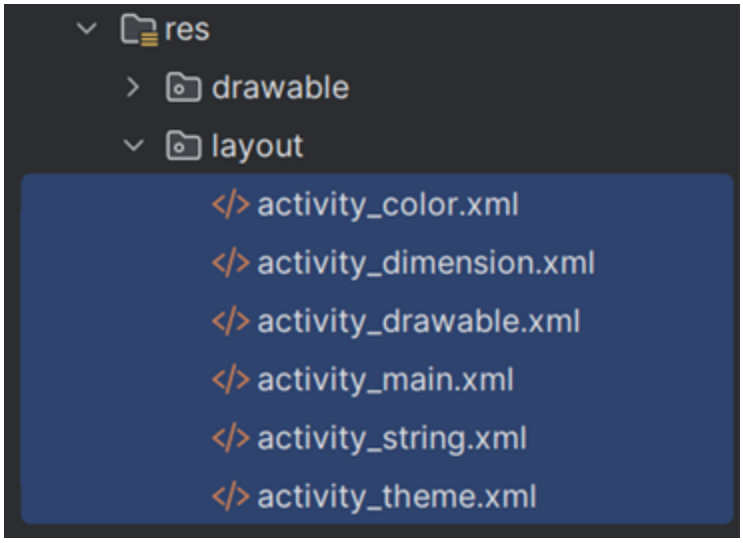


Image Detail: pebbles.jpg

(Copy your image file from your local folder and only copy and paste into drawable folder. Don't Drag and Drop). Image size must within KB size



Total 6 Layout .xml files path



1.Filename: activity_color.xml

```
<?xml version="1.0" encoding="utf-8"?>
 <RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
      android:layout_width="match_parent"
      android:layout_height="match_parent"
      android:background="@color/colorRed">
      <TextView
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Color Activity"
      android:textSize="24sp"
          android:layout_centerInParent="true"
      android:textColor="@color/white"/>
 </RelativeLayout>
```

2.Filename: activity_dimension.xml

```
<?xml version="1.0" encoding="utf-8"?>
 <LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
      android:layout_width="match_parent"
      android:layout_height="match_parent"
      android:orientation="vertical"
      android:padding="16dp"
      android:gravity="center">
      <TextView
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Dimension Activity"
      android:textSize="24sp"
```

```
      android:textColor="@android:color/black"/>
      <!-- Button demonstrating dimension usage -->
      <Button
         android:id="@+id/dimensionButton"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Click Me!"
      android:textSize="@dimen/button_text_size"
      android:padding="@dimen/button_padding"
      android:layout_marginTop="@dimen/button_margin_top"
      android:background="@color/btn_color"/>
 </LinearLayout>
```

3.Filename: activity_drawable.xml

```
<?xml version="1.0" encoding="utf-8"?>
 <RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
      android:layout_width="match_parent"
      android:layout_height="match_parent"
      android:background="@drawable/pebbles">
      <TextView
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Drawable Activity"
      android:textSize="24sp"
      android:layout_centerInParent="true"
      android:textColor="@android:color/white"/>
 </RelativeLayout>
```

4.Filename: activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
 <RelativeLayout
      xmlns:android="http://schemas.android.com/apk/res/android"
      xmlns:tools="http://schemas.android.com/tools"
      android:layout_width="match_parent"
      android:layout_height="match_parent"
      tools:context=".MainActivity"
      android:background="@drawable/pebbles">
      <Button
      android:id="@+id/btnColor"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Color Demo"
      android:layout_centerHorizontal="true"
      android:layout_marginTop="50dp"
      android:onClick="openColorActivity"/>
      <Button
      android:id="@+id/btnTheme"
```

```
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Theme Demo"
      android:layout_below="@id/btnColor"
      android:layout_centerHorizontal="true"
      android:layout_marginTop="20dp"
      android:onClick="openThemeActivity"/>
      <Button
      android:id="@+id/btnString"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="String Demo"
      android:layout_below="@id/btnTheme"
      android:layout_centerHorizontal="true"
      android:layout_marginTop="20dp"
         android:onClick="openStringActivity"/>

      <Button
      android:id="@+id/btnDrawable"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Drawable Demo"
      android:layout_below="@id/btnString"
      android:layout_centerHorizontal="true"
      android:layout_marginTop="20dp"
      android:onClick="openDrawableActivity"/>
      <Button
      android:id="@+id/btnDimension"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Dimension Demo"
      android:layout_below="@id/btnDrawable"
      android:layout_centerHorizontal="true"
      android:layout_marginTop="20dp"
      android:onClick="openDimensionActivity"/>
</RelativeLayout>
```

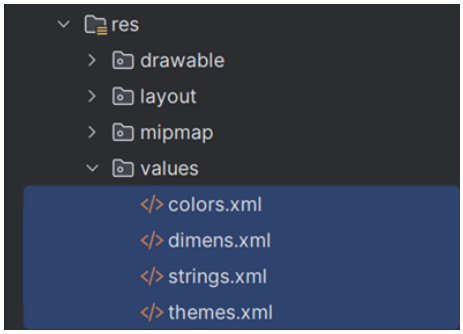
5.Filename: activity_string.xml

```
<?xml version="1.0" encoding="utf-8"?>
 <LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
      android:layout_width="match_parent"
      android:layout_height="match_parent"
      android:orientation="vertical"
      android:gravity="center"
      android:background="@color/colorPrimaryDark">
      <TextView
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="@string/Welcome_Message"
      android:textSize="24sp"
      android:textColor="@android:color/white"/>
 </LinearLayout>
```

6.Filename: activity_theme.xml

```
<?xml version="1.0" encoding="utf-8"?>
 <RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
      android:layout_width="match_parent"
      android:layout_height="match_parent"
      android:background="@android:color/holo_blue_bright">
      <TextView
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="You got this sky blue theme"
      android:textSize="24sp"
      android:layout_centerInParent="true"
      android:textColor="@android:color/black"/>
 </RelativeLayout>
```

Total 4 values .xml files



1.Filename: colors.xml

```
<?xml version="1.0" encoding="utf-8"?>
 <resources>
      <color name="purple_200">#FFBB86FC</color>
      <color name="purple_500">#FF6200EE</color>
      <color name="purple_700">#FF3700B3</color>
      <color name="teal_200">#9ADA03</color>
      <color name="teal_700">#FF018786</color>
      <color name="black">#FF000000</color>
      <color name="white">#FFFFFFFF</color>
      <color name="btn_color">#FFC49662</color>
      <color name="colorRed">#E91E63</color>
      <color name="colorBlue">#0000FF</color>
      <color name="colorPrimary">#6200EE</color>
      <color name="colorPrimaryDark">#3700B3</color>
      <color name="colorAccent">#03DAC5</color>
 </resources>
```

2.Filename: dimens.xml

```
<?xml version="1.0" encoding="utf-8"?>
 <resources>
      <dimen name="button_margin_top">24dp</dimen>
      <dimen name="button_padding">16dp</dimen>
      <dimen name="button_text_size">18sp</dimen>
 </resources>
```

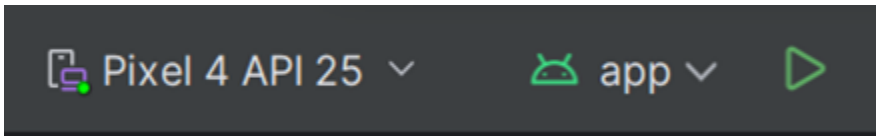
3.Filename: strings.xml

```
<resources>
      <string name="app_name">Practical 2R</string>
      <string name="button_name">Click on me</string>
      <string name="ParaHead">Welcome to Practical 2R</string>
      <string name="Description">This is a string resource demo.</string>
      <string name="Welcome_Message">Welcome to Android Studio. This is your
Practical 2 Resources.</string>
 </resources>
```

4.Filename: themes.xml

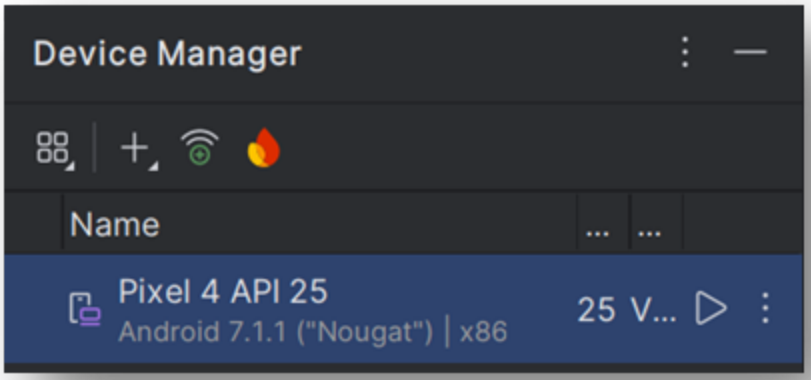
```
<?xml version="1.0" encoding="utf-8"?>
 <resources>
      <style name="AppTheme" parent="Theme.AppCompat.Light.DarkActionBar">
        <item name="colorPrimary">@color/colorPrimary</item>
      <item name="colorPrimaryDark">@color/colorPrimaryDark</item>
      <item name="colorAccent">@color/colorAccent</item>
      </style>
      <style name="BlueTheme" parent="Theme.AppCompat.Light.DarkActionBar">
      <item name="colorPrimary">@color/colorPrimary</item>
      <item name="colorPrimaryDark">@color/colorPrimaryDark</item>
      <item name="colorAccent">@color/colorAccent</item>
      <item name="android:background">#00F0EF</item>
      <item name="android:textColor">#000000</item>
      </style>
 </resources>
```

Above the Android Studio near Main Menu Click the Play button to Run the Program



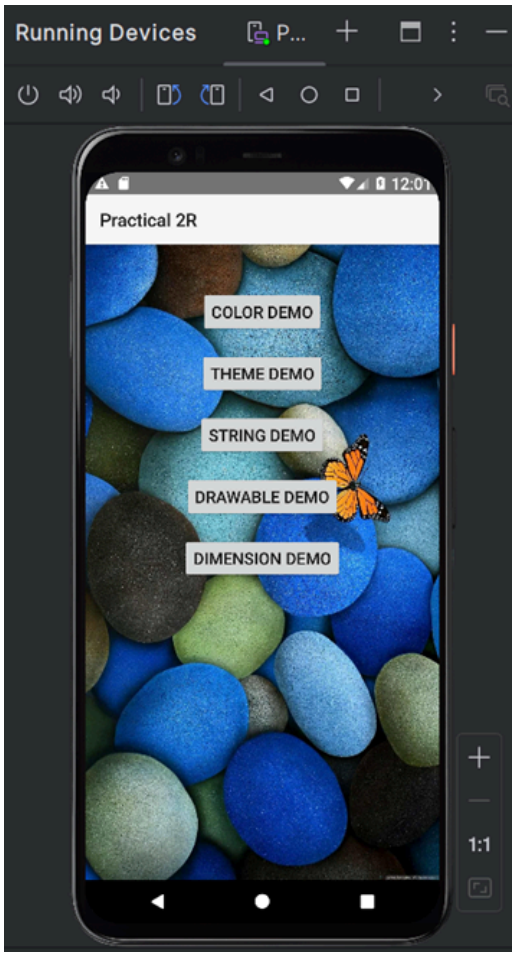
Recommended compulsory AVD Emulator: Pixel 4 API 25 with version 7.1.1 Nougat

Play Emulator

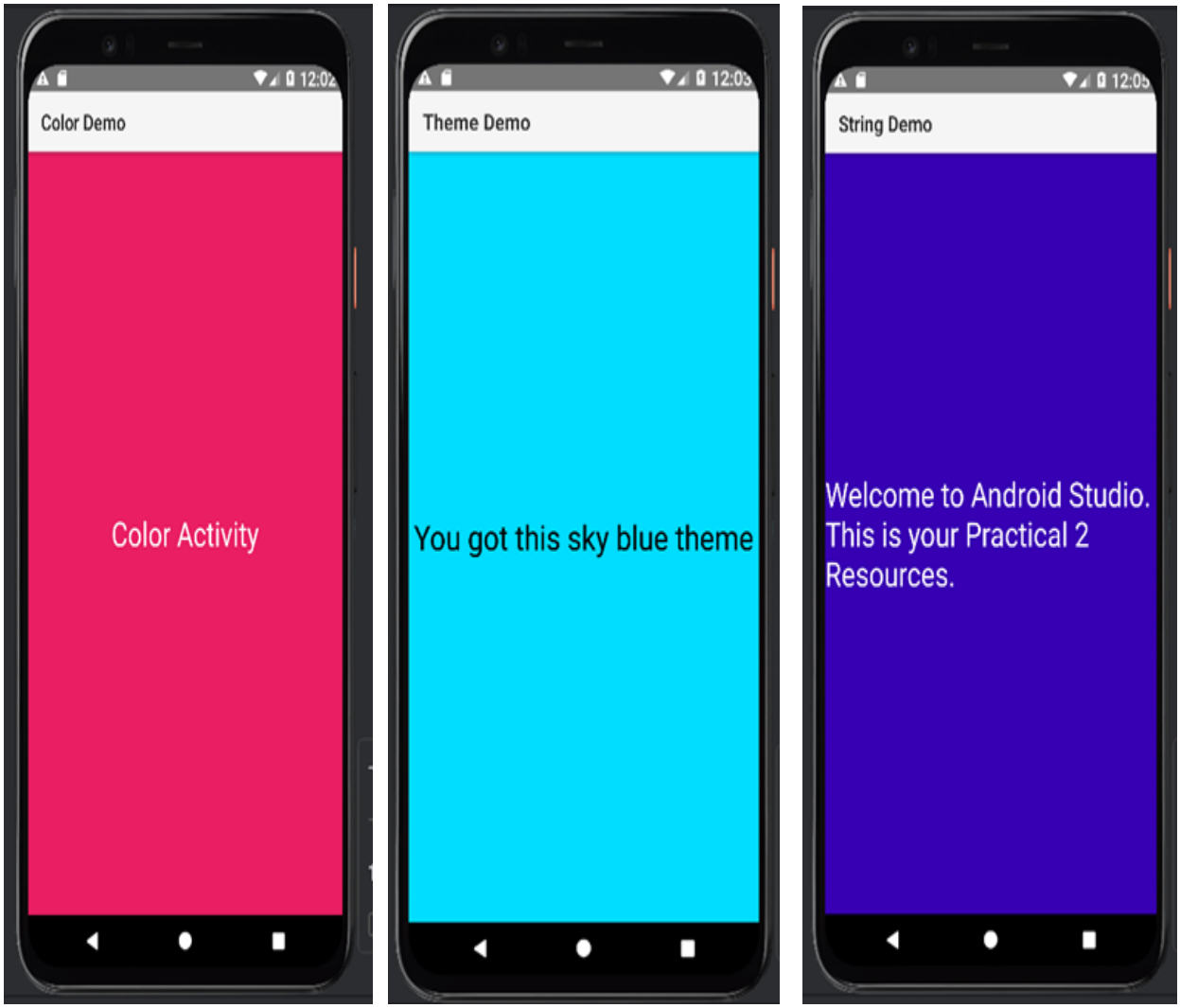


Output:

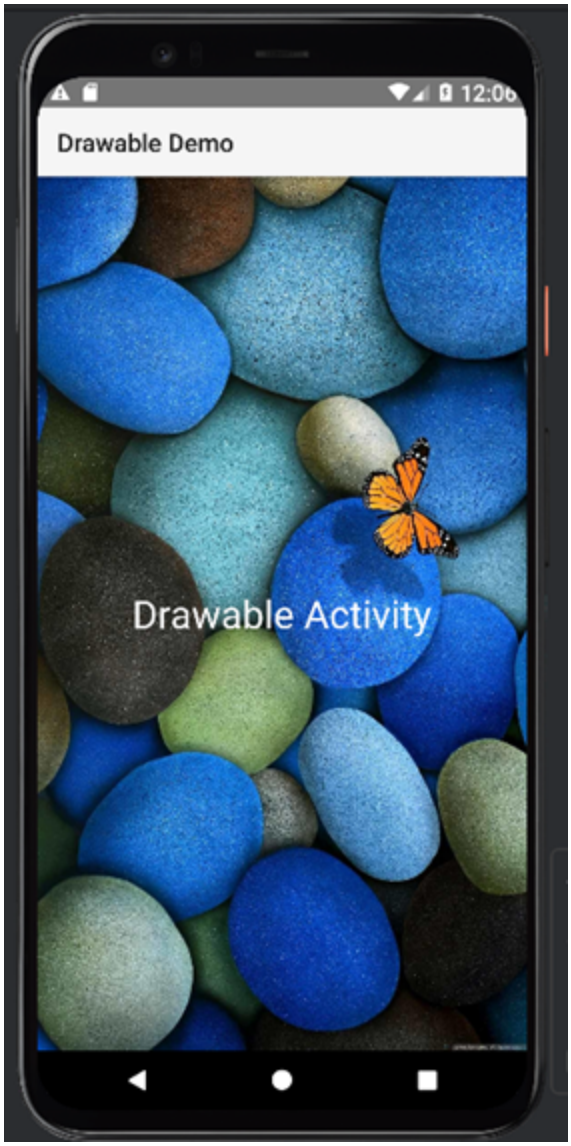
MAIN PAGE



Click COLOR DEMO press back & Click THEME DEMO press back & Click THEME DEMO



Press back & Click COLOR DEMO Press back & Click COLOR DEMO


12:06
Drawable Demo
Drawable Activity


12:06
Dimension Demo
Dimension Activity
CLICK ME!